II. Affine Spaces & Maps

1. Spaces

Def II.1.1. $A \subseteq \mathbb{R}^n$ is called affine iff $\forall x, x' \in A$. the line determined by x & x' is in A

Def II.1.2. An affine combination of $p_0, \dots, p_m$ in $\mathbb{R}^n$ is a point $x := \sum_{k=0}^{m} t_k p_k$.
where $\sum_{k=0}^{m} t_k = 1$. If $t_k \geq 0$ the x is called a convex combination.

Thm II.1.3. If $p_0, \dots, p_m \in \mathbb{R}^n$. then $[p_0, \dots, p_m]$ the convex set spanned by p_k = the set of all convex combinations,

proof. let $S :=$ the set of all convex combinations.

① $[p_0, \dots, p_m] \subseteq S$: It suffices to show that S is a convex set containing p_j.

1°. $p_j = \sum_{k=0}^{n} t_k p_k$. $t_k = 0$ $(k \neq j)$; $t_k = 1$ $(k = j)$ thus $p_k \in S$

2°. $\forall \alpha = \sum_{k=0}^{n} t_k p_k$. $\beta = \sum_{k=0}^{n} s_k p_k$. We have $\forall 0 \leq t \leq 1$, $t\alpha + (1-t)\beta$

$$= \sum_{k=0}^{n} (t \cdot t_k + (1-t) s_k) p_k \quad \text{where} \quad \begin{cases} \sum_{k=0}^{n} t \cdot t_k + (1-t) \cdot s_k = 1 \\ t \cdot t_k + (1-t) s_k \geq 0. \end{cases}$$

thus. $t\alpha + (1-t)\beta \in S \Rightarrow [p_0, \dots, p_m] \subseteq S$

② $S \subseteq [p_0, \dots, p_m]$: For every convex set X containing $\{p_i\}$, we show that $S \subseteq X$.

1°. $m = 0$. then $S = \{p_0\} \subseteq X$

2°. Suppose it's true for m. Consider $m+1$: $S \subseteq [p_0, \dots, p_m]$, X contains $\{p_i\}$.

$\forall t_i \geq 0$. $\sum_{k=0}^{m+1} t_k = 1$. let $\begin{cases} p := \sum_{k=0}^{m+1} t_k p_k \\ t_0 \neq 1 \end{cases}$ $q := \sum_{k=1}^{m+1} \frac{t_k}{1-t_0} p_k$.

By induction: $q \in X$. then $p = t_0 p_0 + (1-t_0) q \in X$ □

Def II.1.4. An ordered set of $\{p_0, \dots, p_m\} \subseteq \mathbb{R}^n$ is called affine independent iff
$\{p_1 - p_0, \dots, p_m - p_0\}$ is linearly independent in vector space $\mathbb{R}^n$

Thm II.1.5. let $\{p_0, \dots, p_m\} \subseteq \mathbb{R}^n$ be an ordered set, then TFAE.

(i) $\{p_0, \dots, p_m\}$ is affine independent

(ii) If $\{s_0, \dots, s_m\} \subseteq \mathbb{R}^{m+1}$ satisfies $\sum_{k=0}^{m} s_k p_k = 0$ and $\sum_{k=0}^{m} s_k = 0$

Then $s_0 = \dots = s_m = 0$

(iii) $\forall x \in [p_0, \dots, p_m]$. $\exists!\ t_0, \dots, t_m \in \mathbb{R}$. s.t. $x = \sum_{k=0}^{m} t_k p_k$, $\sum_{k=0}^{m} t_k = 1$

proof: (i) $\Rightarrow$ (ii). $\{p_1 - p_0, \dots, p_m - p_0\}$ linearly independent in $\mathbb{R}^n$.

and $\sum_{k=0}^{m} s_k (p_k - p_0) = \sum_{k=0}^{m} s_k p_k - (\sum_{k=0}^{m} s_k) p_0 = 0 \Rightarrow s_k = 0$, $k \geq 0$

(ii) $\Rightarrow$ (iii). $x \in [p_0, \dots, p_m]$. then assume that $x = \sum_{k=0}^{m} t_k p_k = \sum_{k=0}^{m} s_k p_k$

then $\sum_{k=0}^{m} (t_k - s_k) p_k = 0$. $\sum_{k=0}^{m} t_k - s_k = 0 \Rightarrow t_k = s_k$, $k \geq 1$

(iii) $\Rightarrow$ (i). let $\sum_{k=0}^{m} s_k (p_k - p_0) = 0$.

$-(s_1 + \dots + s_m) p_0 + s_1 p_1 + \dots + s_m p_m = 0$.

① $s_1 + \dots + s_m \neq 0$. then $p_0 = \sum_{k=1}^{m} \frac{s_k}{s_1 + \dots + s_m} p_k$

By (iii) we have $s_1 = \dots = s_m = 0$ Contradiction!

② $s_1 + \dots + s_m = 0$. then $-(s_2 + \dots + s_m) p_1 + \dots + s_m p_m = 0$. By making induction,

It's easy to have $s_1 = \dots = s_m = 0$ □.

Cora II.1.6. Affine independence is a property of $\{p_0, \dots, p_m\}$ that is independent of the order.

Cora II.1.7. If $A \subseteq \mathbb{R}^n$ is an affine set generated by an affine independence set $\{p_0, \dots, p_m\}$

then $\exists x_0 \in \mathbb{R}^n$. $V \subseteq \mathbb{R}^n$. $\dim V = m$. s.t. $A = V + x_0$

pf: Let $V = \mathrm{Span}(p_1 - p_0, \dots, p_m - p_0)$ $x_0 = p_0$. □.

Def II.1.8. $\{a_1, \dots, a_k\} \subseteq \mathbb{R}^n$ is called in general position. iff $\forall (n+1)$ of its points in forms an affine independent set.

Thm II.1.9. For every $k \geq 0$ $\mathbb{R}^n$ contains k points in general position.

proof. Assume $k > n+1$. Select distinct reals $r_1, \dots, r_k$

let $a_i := (r_i, r_i^2, \dots, r_i^n) \in \mathbb{R}^n$

If $\{a_i\}$ is not in general position.

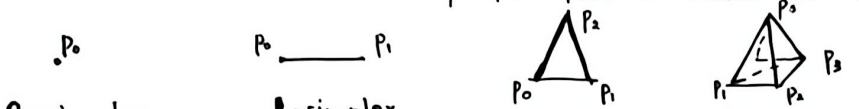
then $\exists (n+1)$ of points W.L.O.G. $\{a_1, \dots, a_{n+1}\}$ are affine dependent.

that is $\{a_2 - a_1, \dots, a_{n+1} - a_1\}$ are linearly dependent.

But $\det \begin{pmatrix} r_2 - r_1 & r_2^2 - r_1^2 & \dots & r_2^n - r_1^n \\ \vdots & \vdots & & \vdots \\ r_{n+1} - r_1 & r_{n+1}^2 - r_1^2 & \dots & r_{n+1}^n - r_1^n \end{pmatrix} = \det \begin{pmatrix} 1 & r_1 & r_1^2 & \dots & r_1^n \\ \vdots & & & & \\ 1 & r_{n+1} & r_{n+1}^2 & \dots & r_{n+1}^n \end{pmatrix}$

$= \prod_{k \leq i < j \leq n+1} (r_j - r_i) \neq 0$ Contradiction □.

Def II.1.10. Let $\{p_1, \dots, p_m\}$ be an affine independent set of $\mathbb{R}^n$. The convex set spanned by this set. ① noted by $[p_0, \dots p_m]$ is called the (affine) m-simplex with vertices p_i



0-simplex 1-simplex 2-simplex 3-simplex ...

Def II.1.11. Let $[p_0, \dots p_m]$ be an m-simplex. The face opposite p_i is $[p_0, \dots, \hat{p_i}, \dots, p_m] := \{\sum t_j p_j : t_j \geq 0, \sum t_j = 1, \text{and } t_i = 0\}$.
The boundary of $[p_0, \dots, p_m]$ is the union of faces.

2. Affine maps.

Def II.2.1 $\{p_0, \dots, p_m\} \subseteq \mathbb{R}^n$ affine independent. A be the affine set it spans
A map $T: A \to \mathbb{R}^k$ is called affine iff $T(\sum t_j p_j) = \sum t_j T(p_j)$
where $\sum t_j = 1$.

Thm II.2.2. $[p_0, \dots, p_m]$ is an m-simplex. $[q_0, \dots, q_n]$ is an n-simplex.
$f: \{p_0, \dots, p_m\} \to [q_0, \dots, q_n]$ be any function. then $\exists!$ affine map
$T: [p_0, \dots, p_m] \to [q_0, \dots, q_n]$ with $T(p_i) = f(p_i)$

proof. Def. $T(\sum t_i p_i) := \sum t_i f(p_i)$. □.